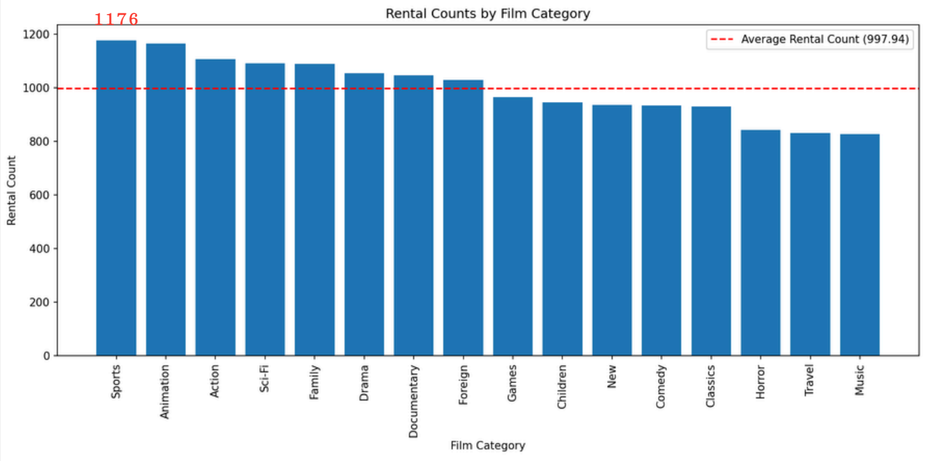
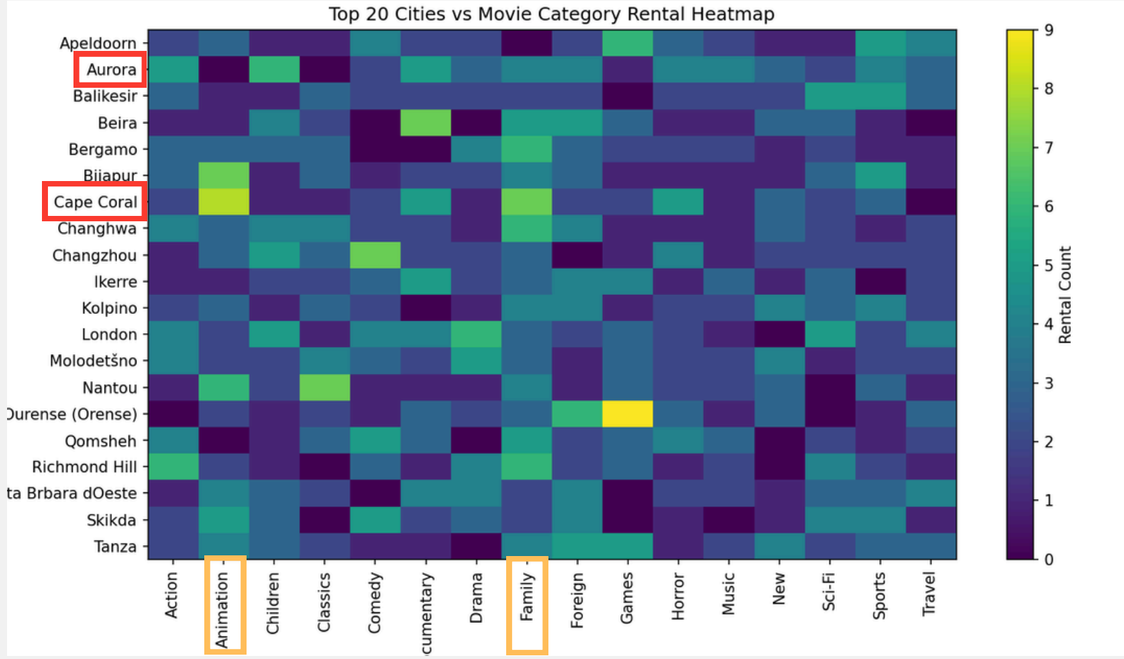
Rental Performance Evaluation Report

Which film genres are the most prevalent?

1. sports cinema
2. cartoon
3. Action films



1 Analysis of Film Genre Preferences in the Top 20 Cities with Elevated Rental Income



Case Study 1: Aurora Favors Urban Areas with Varied and Stable Markets

Map Observation

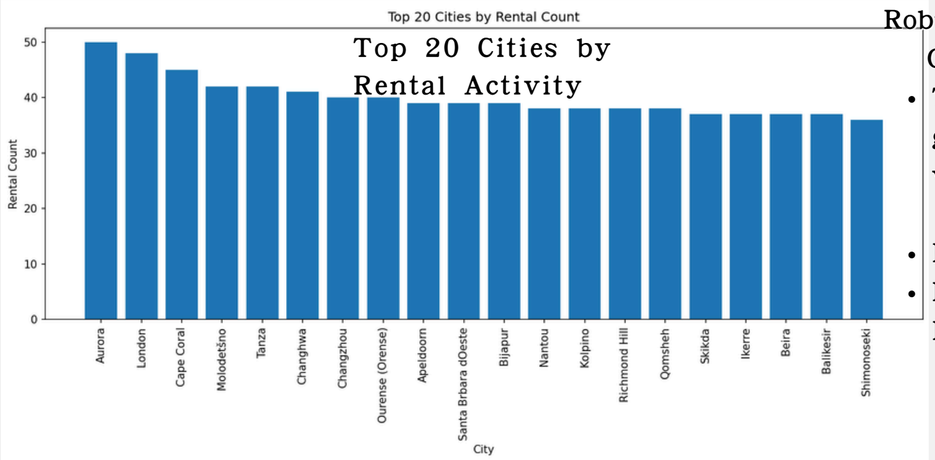
- No single type is especially favored; the overall distribution remains balanced.

Marketing guidance

- New product evaluation
- All forms of promotional activities

Which cities exhibit the highest rental volumes?

1. Aurora, United States
2. London, United Kingdom
3. Cape Collar, United States



2 Global Distribution Map of the Top 2

Case Study 2: Cape Coral - A City with a Distinct and Robust Identity

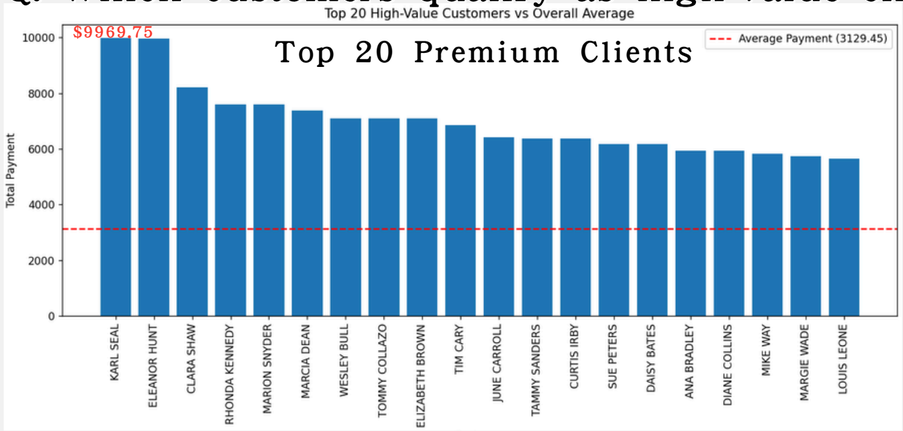
Observation on the Map

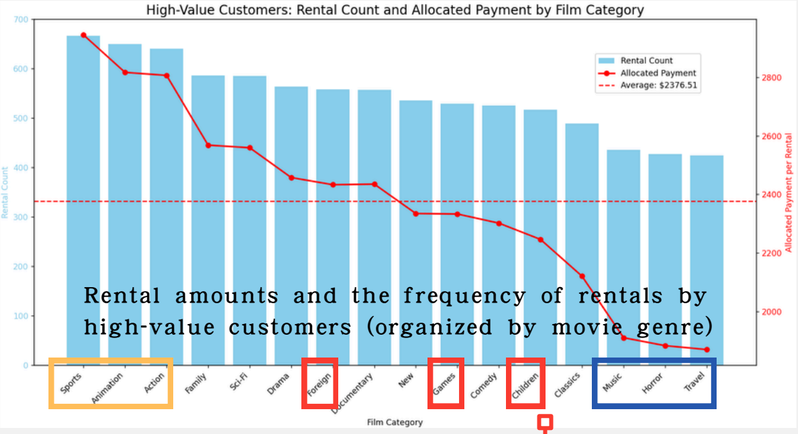
- The Animation / Family genre is notably more vibrant.

Marketing guidance

- Parent-child relationship
- Family Membership Program

Q: Which customers qualify as high-value clients? Which films exhibit low rental rates? Should they be eliminated or promoted

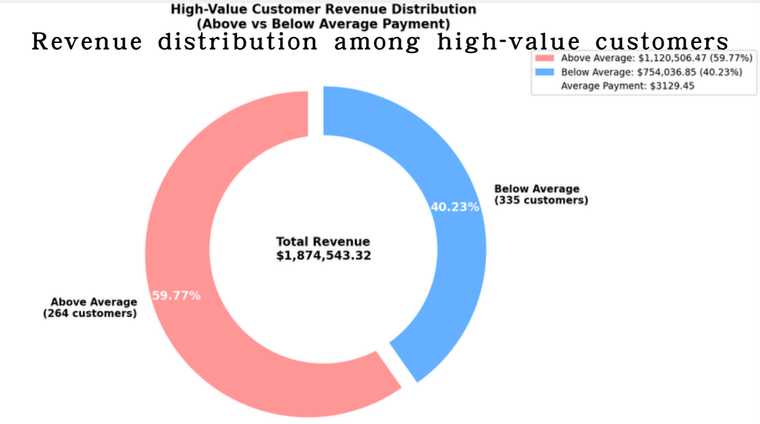


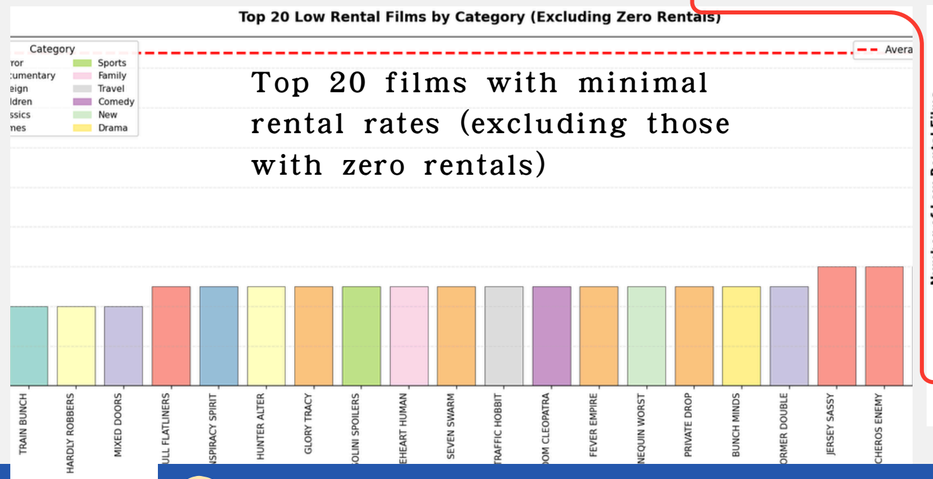


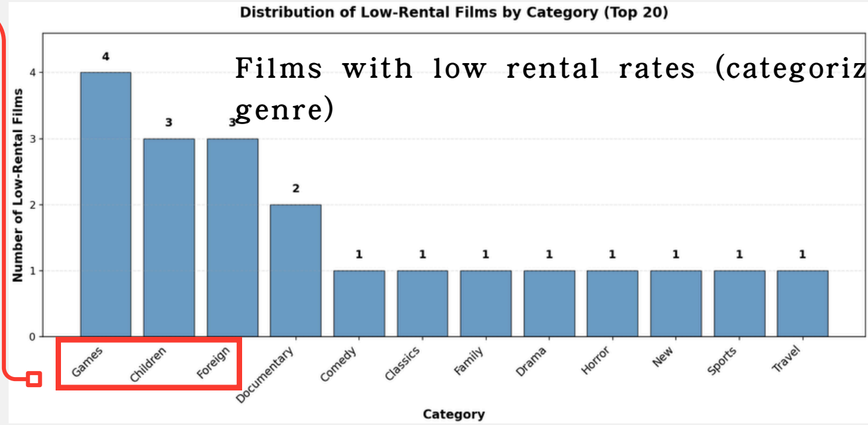
1 Analysis of Film Genre Preferences Among High-Value Customers

Top three video genres preferred by high-value customers:

1. Sports：2944.8
2. Animation：2816.8
3. Action：2806.5







2 Customer Revenue Percentage Assessment

- Definition of high-value clientele:

Customers whose total expenditure surpasses the average customer expenditure ($3129.4)

- Total customers: 599
- Total Revenue: $67,406.56
- High-value customers (264 individuals): $17,817.64
- Revenue generated from high-value customers constituted 59.77% of total revenue.

3 Recommendations for the removal of videos from streaming platforms and promotional initiatives.

- Films with no rentals should be removed from shelves to minimize sales expenses. (Appendix 8)
- Prioritize the elimination or significant promotion of: Games, Children, and Foreign films. These three categories contain the highest number of films with low rental rates (3-4 films) but rank lower in popularity and exhibit reduced spending among high-value customers, suggesting inadequate market demand.
- Suggestions for the promotion or replacement of content: Music, Horror, Travel → Allocate resources to more lucrative series.

Appendix 1: The most popular film genres (sorted by rental frequency)

SQL query statement

```
# === 2. SQL 查詢（影片類型） ===
query = """
SELECT
    c.name AS 影片類型,
    COUNT(r.rental_id) AS 租借次數
FROM rental r
JOIN inventory i ON r.inventory_id = i.inventory_id
JOIN film f ON i.film_id = f.film_id
JOIN film_category fc ON f.film_id = fc.film_id
JOIN category c ON fc.category_id = c.category_id
GROUP BY c.name
ORDER BY 租借次數 DESC;
"""
cursor.execute(query)
```

```
# 計算平均租借次數
avg_rental = df["租借次數"].mean()
print("整體平均租借次數：", avg_rental)
```

```
# 加上「整體平均租借次數」水平線
plt.axhline(
    y=avg_rental,
    color="red",
    linestyle="--",
    label=f"Average Rental Count ({avg_rental:.2f})"
)
plt.legend()
plt.tight_layout()
plt.show()
```

Export outcomes

	影片類型	租借次數
▶	Sports	1176
	Animation	1164
	Action	1107
	Sci-Fi	1092
	Family	1089
	Drama	1055
	Documentary	1047
	Foreign	1028

Appendix 2: Cities with the Highest Rental Volumes (Statistics Organized by City)

SQL query statement

```
# === 2. SQL 查詢（城市 + 國家 + 租借次數）===
query = """
SELECT
    ci.city AS 城市,
    co.country AS 國家,
    COUNT(r.rental_id) AS 租借次數
FROM rental r
JOIN customer cu ON r.customer_id = cu.customer_id
JOIN address a ON cu.address_id = a.address_id
JOIN city ci ON a.city_id = ci.city_id
JOIN country co ON ci.country_id = co.country_id
GROUP BY ci.city, co.country
ORDER BY 租借次數 DESC;
"""
```

Export outcomes

城市	租借次數
Aurora	50
London	48
Cape Coral	45
Molodetšno	42
Tanza	42
Changhwa	41
Changzhou	40
Ourense (Orense)	40
Apeldoorn	39

Global urban distribution map

```
# === 5. Top 20 城市 → 經緯度（只做 20 筆！）===
from geopy.geocoders import Nominatim
import time
from geopy.exc import GeocoderTimedOut, GeocoderServiceError
geolocator = Nominatim(user_agent="sakila-city-map", timeout=10)

def geocode(city):
    try:
        location = geolocator.geocode(city)
        time.sleep(1)  # 避免過快請求
        if location:
            return pd.Series([location.latitude, location.longitude])
    except (GeocoderTimedOut, GeocoderServiceError):
        print(f"Geocode timeout: {city}")
    return pd.Series([None, None])

top20[["lat", "lon"]] = top20["城市"].apply(geocode)
top20 = top20.dropna()
```

```
# === 6. Plotly 城市世界地圖（Top 20）===
fig = px.scatter_geo(
    top20,
    lat="lat",
    lon="lon",
    size="租借次數",
    color="租借次數",
    hover_name="城市",
    projection="natural earth",
    title="Top 20 Cities by Rental Count",
    size_max=40
)
fig.show()
```

Appendix 3: Evaluation of film rental performance in the top 20 rental cities

SQL query statement

```
# === 2. SQL 查詢 ===
query = """
SELECT
    ci.city AS 城市,
    ca.name AS 影片類型,
    COUNT(r.rental_id) AS 租借次數
FROM rental r
JOIN inventory i ON r.inventory_id = i.inventory_id
JOIN film f ON i.film_id = f.film_id
JOIN film_category fc ON f.film_id = fc.film_id
JOIN category ca ON fc.category_id = ca.category_id
JOIN customer cu ON r.customer_id = cu.customer_id
JOIN address a ON cu.address_id = a.address_id
JOIN city ci ON a.city_id = ci.city_id
GROUP BY ci.city, ca.name;
"""

df = pd.read_sql(query, engine)
# === 3. Top 20 城市 ===
top20_cities = (
    df.groupby("城市")["租借次數"]
    .sum()
    .sort_values(ascending=False)
    .head(20)
    .index
)

df_top20 = df[df["城市"].isin(top20_cities)]

# === 4. 各城市最受歡迎影片類型 ===
city_favorite_category = (
    df_top20
    .sort_values(["城市", "租借次數"], ascending=[True, False])
    .groupby("城市")
    .head(1)
    .reset_index(drop=True)
)
print(city_favorite_category)
```

Export outcomes

城市	影片類型	租借次數
Araçatuba	Action	2
Vila Velha	Action	3
Kurashiki	Action	2
Poços de Caldas	Action	3
Warren	Action	2
Città del Vaticano	Action	3
La Plata	Action	4
Denizli	Action	1

城市	影片類型	租借次數
Cianjur	Animation	7
Kitwe	Animation	4
Bandar Seri Beg...	Animation	1
Alessandria	Animation	2
Ezhou	Animation	4
Purnea (Purnia)	Animation	2
Sirjan	Animation	1

Visualized Heatmap (City × Film Genre)

```
# === 5. Heatmap ===
pivot = df_top20.pivot_table(
    index="城市",
    columns="影片類型",
    values="租借次數",
    aggfunc="sum",
    fill_value=0
)

plt.figure(figsize=(14, 8))
plt.imshow(pivot, aspect="auto")
plt.colorbar(label="Rental Count")
plt.xticks(range(len(pivot.columns)), pivot.columns, rotation=90)
plt.yticks(range(len(pivot.index)), pivot.index)
plt.title("Top 20 Cities vs Movie Category Rental Heatmap")
plt.tight_layout()
```

Appendix 4: Twenty Most Valuable Customers

SQL query statement

```
# 查詢高價值顧客的總付款金額與租借次數
query = """
SELECT
    c.customer_id,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    COUNT(r.rental_id) AS total_rentals,
    SUM(p.amount) AS total_payment
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
JOIN payment p ON c.customer_id = p.customer_id
GROUP BY c.customer_id
HAVING
    -- 篩選條件：僅保留總支付金額高於平均值的客戶
    SUM(p.amount) > (
        SELECT AVG(total_amount) -- 計算子查詢中所有客戶的平均支付金額
        FROM (
            SELECT SUM(amount) AS total_amount -- 計算每位客戶的總支付金額
            FROM payment
            GROUP BY customer_id
        ) AS avg_table
    )
ORDER BY total_payment DESC;
"""
```

Export outcomes

customer_id	customer_name	total_rentals	total_payment
526	KARL SEAL	2025	9969.75
148	ELEANOR HUNT	2116	9960.84
144	CLARA SHAW	1764	8214.36
137	RHONDA KENNEDY	1521	7589.79
178	MARION SNYDER	1521	7589.79
236	MARCIA DEAN	1764	7374.36
469	WESLEY BULL	1600	7104.00

Appendix 5: Rental Amount and Frequency of Premium Customers (Ranked by Film Genre)

SQL query statement

```
# 高價值顧客在各類別的租借次數與分攤金額
query = """
WITH customer_total AS (
    -- 計算每位顧客的「終身消費總額（Lifetime Value）」
    SELECT
        customer_id,
        SUM(amount) AS total_amount
    FROM payment
    GROUP BY customer_id
),

avg_all AS (
    -- 計算所有顧客終身消費總額的「平均值」
    --     作為判斷高價值顧客的門檻
    SELECT
        AVG(total_amount) AS avg_lifetime_value
    FROM customer_total
),

high_value_customers AS (
    -- 篩選高價值顧客（消費總額高於全體平均）
    --     同時計算其租借次數，用於後續金額分攤
    SELECT
        ct.customer_id,
        ct.total_amount AS total_payment,
        COUNT(r.rental_id) AS total_rentals
    FROM customer_total ct
    -- 使用 CROSS JOIN 取得全體平均值（單一列）
    CROSS JOIN avg_all
```

```
    -- LEFT JOIN 確保即使租借次數為 0 也不會被排除
    LEFT JOIN rental r
        ON ct.customer_id = r.customer_id
    WHERE ct.total_amount > avg_all.avg_lifetime_value
    GROUP BY ct.customer_id, ct.total_amount
)

SELECT
    -- 以影片類型為分析單位
    cat.name AS category_name,

    -- 高價值顧客在該類型的租借次數
    COUNT(r.rental_id) AS rental_count,

    -- 將高價值顧客的總消費金額平均分攤至其租借次數
    --     避免因 JOIN rental 而重複累加付款金額
    --     NULLIF 用於避免除以 0 的情況
    SUM(
        hvc.total_payment / NULLIF(hvc.total_rentals, 0)
    ) AS allocated_payment

FROM rental r
-- 只分析高價值顧客的租借行為
JOIN high_value_customers hvc
    ON r.customer_id = hvc.customer_id
JOIN inventory i
    ON r.inventory_id = i.inventory_id
JOIN film f
    ON i.film_id = f.film_id
JOIN film_category fc
    ON f.film_id = fc.film_id
JOIN category cat
    ON fc.category_id = cat.category_id

GROUP BY cat.name
ORDER BY rental_count DESC;
"""
```

Export outcomes

category_name	rental_count	allocated_payment
Sports	667	2944.790319
Animation	650	2816.776499
Action	641	2806.474074
Family	586	2568.316847
Sci-Fi	585	2559.519056
Drama	564	2457.672332

Appendix 6: Revenue Proportion of High-Value Customers

SQL query statement

```
# 查詢所有顧客的總付款金額與租借次數
query = """
SELECT
    c.customer_id,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    COUNT(r.rental_id) AS total_rentals,
    SUM(p.amount) AS total_payment
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
JOIN payment p ON c.customer_id = p.customer_id
GROUP BY c.customer_id
ORDER BY total_payment DESC;
"""
```

Python computes the data required for a pie chart

```
# 計算平均消費金額
avg_payment = df["total_payment"].mean()
print("整體平均消費金額：", avg_payment)

# 區分高價值與一般顧客
df['customer_segment'] = df['total_payment'].apply(
    lambda x: 'Above Average' if x > avg_payment else 'Below Average'
)

above_avg_customers = df[df['customer_segment'] == 'Above Average']
below_avg_customers = df[df['customer_segment'] == 'Below Average']

# 統計數據
total_revenue = df['total_payment'].sum()
above_avg_count = len(above_avg_customers)
below_avg_count = len(below_avg_customers)
above_avg_revenue = above_avg_customers['total_payment'].sum()
below_avg_revenue = below_avg_customers['total_payment'].sum()
above_avg_revenue_pct = (above_avg_revenue / total_revenue) * 100
below_avg_revenue_pct = (below_avg_revenue / total_revenue) * 100
```

Export outcomes

customer_id	customer_name	total_rentals	total_payment
526	KARL SEAL	2025	9969.75
148	ELEANOR HUNT	2116	9960.84
144	CLARA SHAW	1764	8214.36
137	RHONDA KENNEDY	1521	7589.79
178	MARION SNYDER	1521	7589.79
236	MARCIA DEAN	1764	7374.36

Appendix 7: Twenty Movies with the Lowest Rental Rates (excluding those with zero rentals)

SQL query statement

```
# 查詢低於平均租賃次數的前20部電影
query = """
-- 查詢計算每部電影的租賃次數
WITH film_rentals AS (
    SELECT
        f.film_id,
        f.title,
        c.name AS category_name,
        COUNT(r.rental_id) AS rental_count
    FROM film f
    JOIN film_category fc ON f.film_id = fc.film_id
    JOIN category c ON fc.category_id = c.category_id
    JOIN inventory i ON f.film_id = i.film_id
    LEFT JOIN rental r ON i.inventory_id = r.inventory_id
    GROUP BY f.film_id, f.title, c.name
    HAVING rental_count > 0  -- 只選擇租賃次數大於0的電影
),
-- 計算所有電影的平均租賃次數
avg_rental AS (
    SELECT AVG(rental_count) AS avg_rental_count
    FROM film_rentals
)
SELECT
    fr.film_id,
    fr.title,
    fr.category_name,
    fr.rental_count,
    ar.avg_rental_count
FROM film_rentals fr
CROSS JOIN avg_rental ar
WHERE fr.rental_count < ar.avg_rental_count -- 只選擇租賃次數低於平均值的電影
ORDER BY fr.rental_count ASC
LIMIT 20; -- 只顯示前20條記錄
```

Export outcomes

film_id	title	category_name	rental_count	avg_rental_count
904	TRAIN BUNCH	Horror	4	16.7589
400	HARDLY ROBBERS	Documentary	4	16.7589
584	MIXED DOORS	Foreign	4	16.7589
343	FULL FLATLINERS	Children	5	16.7589
180	CONSPIRACY SPIRIT	Classics	5	16.7589

Appendix 8: Films with Minimal Rental Rates (Categorized by Film Genre)

SQL query statement

```
# 查詢低租借率影片及建議
query = """
USE sakila;
WITH film_rentals AS (
    SELECT
        f.film_id,
        f.title,
        c.name AS category_name,
        COUNT(r.rental_id) AS rental_count -- 計算每部電影的租借次數
    FROM film f
    JOIN film_category fc ON f.film_id = fc.film_id
    JOIN category c ON fc.category_id = c.category_id
    LEFT JOIN inventory i ON f.film_id = i.film_id
    LEFT JOIN rental r ON i.inventory_id = r.inventory_id
    GROUP BY f.film_id, f.title, c.name -- 按電影 ID、標題和類別名稱分組
),
avg_rental AS (
    SELECT AVG(rental_count) AS avg_rental_count -- 計算所有電影的平均租借次數
    FROM film_rentals
)
SELECT
    ROW_NUMBER() OVER (ORDER BY fr.rental_count ASC) AS `row_number`,
    fr.film_id,
    fr.title,
    fr.category_name,
    fr.rental_count,
    ar.avg_rental_count,
    -- 根據租借次數提供建議
    CASE
        WHEN fr.rental_count = 0 THEN '下架建議' -- 如果租借次數為 0，建議下架
        WHEN fr.rental_count < ar.avg_rental_count THEN '促銷建議' -- 如果租借次數低於平均值，建議促銷
        ELSE '正常' -- 否則標記為正常
    END AS suggestion
FROM film_rentals fr
CROSS JOIN avg_rental ar
ORDER BY fr.rental_count ASC;
```

Export outcomes

row_number	film_id	title	category_name	rental_count	avg_rental_count	suggestion
36	195	CROWDS TELEMARK	Sci-Fi	0	16.0550	下架建議
37	860	SUICIDES SILENCE	Sci-Fi	0	16.0550	下架建議
38	671	PERDITION FARGO	Sports	0	16.0550	下架建議
39	41	ARSENIC INDEPEND...	Travel	0	16.0550	下架建議
40	87	BOONDOCK BALLRO...	Travel	0	16.0550	下架建議
41	386	GUMP DATE	Travel	0	16.0550	下架建議
42	642	ORDER BETRAYED	Travel	0	16.0550	下架建議
43	400	HARDLY ROBBERS	Documentary	4	16.0550	促銷建議
44	584	MIXED DOORS	Foreign	4	16.0550	促銷建議
45	904	TRAIN BUNCH	Horror	4	16.0550	促銷建議
46	343	FULL FLATLINERS	Children	5	16.0550	促銷建議

A total of 42 films with no rentals have been recommended for removal from the platform.